CODESPRINT11

ENVISION SUBMISSION

ALGORITHM BATTLE ARENA

Team Number	***CS11-197***
Team Name	*Pathfinders*
University	*Sri Lanka Institute of Information Technology*
Project Subtrack	*Track 01 - Platforms*

Informatics Institute of Technology IEEE Student Branch
IEEE COMPUTER SOCIETY
Informatics Institute of Technology IEEE Student Branch A Student Chapter of the IEEE Robotics & Automation Society
Informatics Institute of Technology IEEE Student Branch
Informatics Institute of Technology IEEE Student Branch A Student Chapter of the IEEE Computational Intelligence Society

1. PROBLEM DEFINITION & BACKGROUND

1.1. Problem Statement

The problem statement should clearly define the issue your solution aims to solve. Keep it concise and impactful, ideally within one or two sentences. Ensure it highlights the problem's urgency, relevance and potential impact on users or industries.

1.2. Problem Background

Provide context on why this problem is significant by discussing its relevance in today's world. Explain how this problem affects individuals, businesses or society. Mention any ongoing challenges or inefficiencies caused by the issue. If required, include statistics, real-world examples, or case studies to validate the significance of the problem.

1.3. Research Background

Mention previous research, case studies, or expert insights that support your problem definition. Use credible sources such as academic papers, industry reports or market research to validate the significance of the problem. Provide citations for any referenced data.

1.4. In-Scope & Out-Scope

In-Scope: *Clearly define what aspects your solution will cover. Specify the functionalities and areas that your project will focus on. This helps set expectations for what will be included in the final solution.*

Out-Scope: *Define what your solution will not cover to avoid confusion. Highlight any assumptions or limitations that must be considered when evaluating your proposal.*

1.5. User Pain Points

Identify the key issues users face due to the chosen problem. Explain the difficulties they encounter and why existing solutions are not addressing them effectively. Highlight unmet needs and gaps in the market that your solution aims to fill.

2. Proposed Solution

2.1. One-Line Pitch

Provide a compelling one-liner summarizing your solution's core value proposition. This should be clear, impactful and easily understandable, similar to an elevator pitch.

2.2. Solution Overview

Describe your proposed solution in detail. Explain what it is, how it works, and what key functionalities it includes. Provide a brief technical overview if applicable.

2.3. Purpose & Impact

Explain the primary goal of your solution. Discuss how it solves the problem and improves user experience. Mention any expected benefits, such as increased efficiency, cost reduction, or enhanced accessibility.

2.4. Ethical & Social Impact

Discuss any ethical considerations your solution addresses, such as data privacy, inclusivity, or environmental impact. Explain how your solution contributes to positive social change or aligns with sustainability goals.

2.5. Uniqueness & Special Features

Highlight what differentiates your solution from existing ones. Discuss any unique technical innovations, proprietary methods, or creative aspects that give your product a competitive edge.

3. Market & Industry Analysis

3.1. Target Audience & Market Size

Define your primary and secondary user segments. Describe their characteristics, needs, and behaviors. Provide estimates of the potential market size.

3.2. Market Potential & Importance

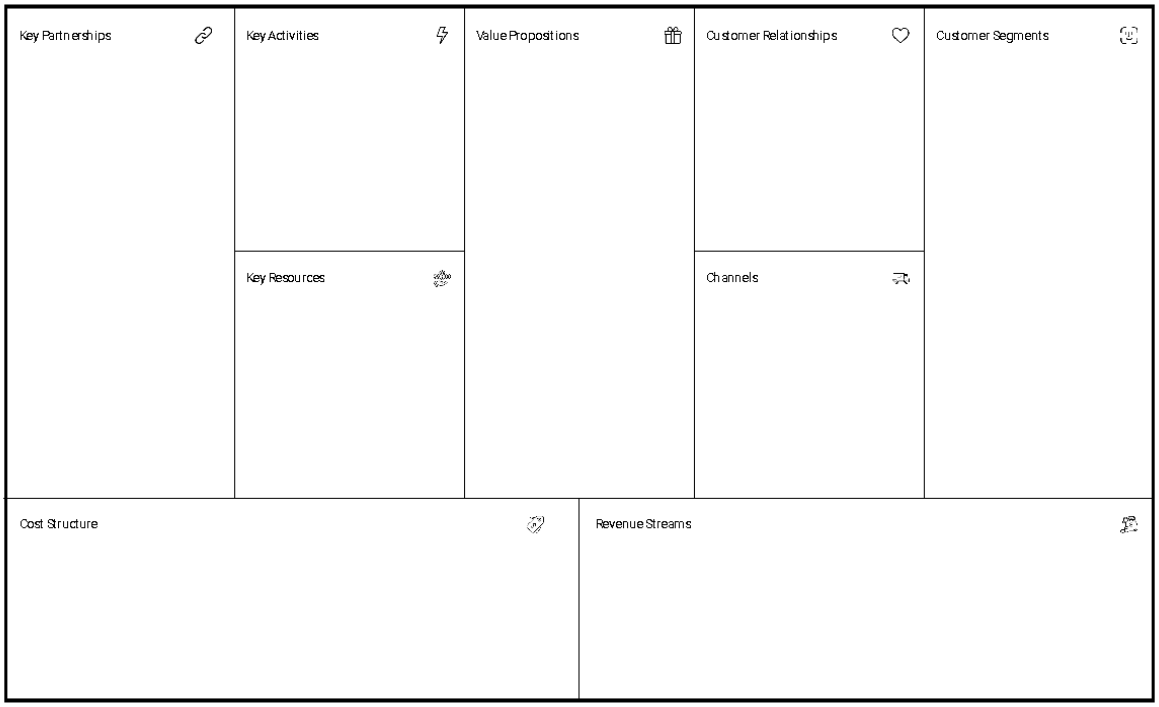
Explain why your solution is relevant in today's market. Discuss industry trends and opportunities that support the need for your product. Highlight the demand and long-term sustainability of your idea along with growth opportunities.

4. Business & Financial Model

4.1. Business Revenue Model & Cost Structure

Explain how your solution will generate revenue. Define your monetization strategy, such as subscriptions, one-time purchases, or advertising. Include a Business Model Canvas highlighting the key aspects of your financial plan.

The Business Model Canvas



Link : https://codesprint.lk/Business-Model-Canvas

4.2. Initial Go-to-Market Strategy

Outline how you plan to introduce your solution to the market. Discuss early adoption strategies, potential pilot programs, and marketing tactics. Mention any strategic partnerships that could help accelerate growth.

4.3. Partnerships & Funding Strategies

Mention potential investors, grants, or revenue sources you plan to explore. Discuss how you will secure funding for development, marketing, and scaling.

5. Implementation

5.1. Tech Stack (Overview)

Mention the key technologies, frameworks, and tools that will be used to build your solution, and provide a brief explanation of how each contributes to its functionality in a few sentences.

5.2. Impact on Stakeholders

Identify the main stakeholders involved (ex: customers, investors, society) and explain how they will benefit from your solution.

5.3. Onion Diagram

Provide an Onion Diagram to visually represent the different layers of stakeholders involved in the project. The diagram should illustrate how various groups (such as core users, supporting stakeholders, and external influences) are positioned based on their level of interaction and impact on the solution. This helps clarify relationships, dependencies, and the overall ecosystem surrounding the project.

5.4. Real-World Scenarios & User Journey

Describe a typical user's interaction with your product. Outline key steps and touchpoints that define the user experience. If applicable, include diagrams or user flow charts.

6. Project Management

6.1. Project Management Approach

Explain the project management methodology that will guide the development process (e.g., Agile, Scrum, or Waterfall). Describe how the team will organize tasks, collaborate, and track progress to ensure the project is completed efficiently.

6.2. Project Timeline & Milestones

Provide a structured timeline outlining the key phases of the project, major milestones, and expected deliverables. This should demonstrate how the development of the Minimum Viable Product (MVP) will progress from planning and development to testing and final implementation.

6.3. Team Roles & Responsibilities

Clearly define the roles and responsibilities of each team member involved in the project. Explain how tasks will be distributed among the team to ensure effective collaboration, accountability, and successful completion of the project

7. Additional and Supportive Materials (Optional)

Include any supplementary materials that will strengthen the submission. These may include:

- *Wireframes & UI/UX Designs – Sketches or high-fidelity designs that illustrate the user experience.*
- *Research Data & Feasibility Reports – Supporting studies, analytics, or surveys that validate the solution.*
- *Diagrams & Flowcharts – Visual representations of workflows, system architecture, or process flows.*
- *Prototypes or MVP Demonstrations – A working model, interactive prototype, or demo video showcasing functionality.*
- *Testimonials or User Feedback – Early insights from potential users or stakeholders.*

Please zip all relevant files and submit them in the 'Additional and Supportive Materials (Optional)' section of the submission form.

8. Team

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Member 1 Name -
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Member 4 Name -
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